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PALEOMAGNETIC EVIDENCE OF LARGE FAULT DISPLACEMENT AROUND THE PO-BASIN

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ABSTRACT

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New paleomagnetic data are presented from Tuscany and the Southern Alps (Italy). A paleomagnetic study of the autochthonous Verrucano near Siena revealed two characteristic paleomagnetic directions, one of the Ladinian to Carnian Verrucano (D = 301.1°, I = +60.2°, α_{95} = 6.0°) and the other of the Late Carboniferous to Early Permian Farma formation (D = 318.8°, I = +20.8°, α_{95} = 12.8°). A paleomagnetic study on a limestone sequence in the Vicentinian Alps revealed characteristic directions for Late Jurassic (D = 328.5°, I = +45.3°, α_{95} = 3.0°), for Turonian—Coniacian (D = 345.5°, I = +38.2°, α_{95} = 3.3°), for Early Santonian (D = 344.2°, I = +43.3°, α_{95} = 5.3°) and for Late Santonian to Early Campanian rocks (D = 342.7°, I = +50.9°, α_{95} = 5.3°). Comparison of these new data with published data of Permian and younger age from the Southern Alps and from Umbria respectively, leads to conclusions about movements of crustal blocks North and South of the Po-basin. Further comparison of the data from the Northern Apennines with published Sardinian data seems to support that Sardinia belonged to the central Italian Peninsula until present times. It is argued that the Northern Apennines and the Vicentinian Alps belonged to the African plate and that their movements were defined by the opening history of the Atlantic. It was found, however, that during Late Cretaceous and again in Early Tertiary the two Italian crustal blocks must have been uncoupled to some degree. This supports the hypothesis of a major fault zone in the area of the Po-basin, which is now buried under a thick Late Tertiary to Quaternary sedimentary cover.

INTRODUCTION

The Po-basin, as it appears today, is an elongated plain between two entirely different fold belts of Alpine origin: the Alps in the North and the Apennines in the South. It is drained by the river Po and its tributaries, and the abundant water supply from the Alps and the Apennines causes rapid accumulation of terrigenous material in the plain. In the Northwest, the Po-

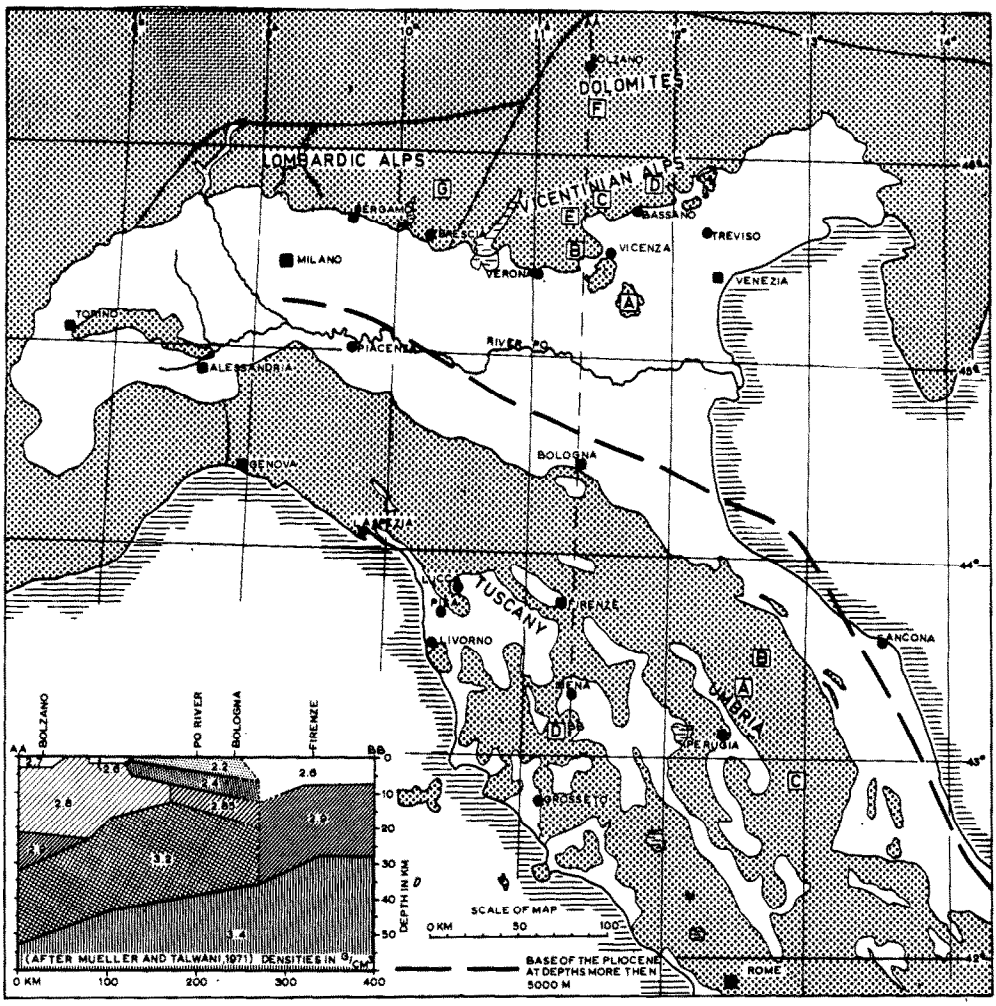


Fig. 1. Map of Northern Italy. Blank areas represent post-Middle Miocene deposits in the Tuscan—Umbrian area, and post-Upper Miocene sediments in the Po plain. The North—South profile is based on gravity and seismic refraction data according to Mueller and Talwani (1971). Dashed line gives location of the Pliocene through axis. The letters are referred to in Lists 1 and 2.

basin is bordered by the most internal massif of the Alps and in the Northeast by the calcareous Bergamask and Vicentinian Alps. The latter can be considered as the western continuation of the Dinarides mountain chain in Yugoslavia. They essentially consist of Mesozoic and Early Tertiary rocks, forming a steep mountain belt, which emerges rather suddenly from the flat basin. The southern margin of the Po-basin is characterized by a low moun-

tain belt containing N—NE dipping Tertiary deposits, which have been invaded, as late as Pliocene times, by olistoliths and olistostromes containing Mesozoic elements (Lucchetti et al., 1962). Subsurface investigations in the Po-basin revealed these allochthonous elements to be present some tens of kilometers North of the present basin margin (Agip Mineraria, 1959). In the North, on the contrary, an enormously thick, undisturbed series of clastics, ranging from Oligocene to Recent times, border the Alps. The Mesozoic and Early Tertiary sediments of the Alps dip steeply southward at this basin margin, suggesting normal faulting towards the South continuing until rather recent times.

The thickness of the authochthonous basin fill is enormous, as is shown by the asymmetric axis in Fig. 1, below which the base of the Pliocene reaches depths of more than 5000 meters (Ogniben, 1973). The considerable thickness of the Tertiary deposits in the Northern Apennines suggests a total thickness of Cenozoic sediments in the basin of some tens of kilometres. It is not surprising that little is known about the Mesozoic sediments which are supposed to be present at still greater depths. On the other hand, seismic studies indicate a normal crustal thickness of 30—35 km (Caloi, 1958; Giese and Morelli, 1973).

The gravity pattern and seismic refraction data indicate a major E—W striking fault zone, which is buried now underneath the thick sedimentary cover (Fig. 1; Mueller and Talwani, 1971). Our study supports the evidence for this fault zone and the relevant paleomagnetic data form the subject of this paper.

Evidence for large-scale crustal movements was previously found in the results from several paleomagnetic studies. The paleomagnetic directions of Permo-Triassic and Eocene rocks from the Southern Alps deviate systematically from the European directions for the same periods. Even North of the Insubric line, which is generally considered to be the fundamental boundary between the Southern Alps and the Central Alps, such deviating directions were found (Förster et al., 1975). On the other hand the paleomagnetic poles from the Southern Alps are very close to the African paleomagnetic poles (Zijderveld and Van der Voo, 1973). These authors concluded that the Southern Alps could have belonged to the African plate during their post-Hercynian evolution. However, this hypothesis must be refined, since it was found that the Northern Apennines also did belong to the African plate, but only until the Early Tertiary; afterwards an independent rotation occurred relative to the African plate (VandenBerg et al., 1976).

Direct comparison of the scanty available data from the Southern Alps and from the Northern Apennines is difficult. The data from the Southern Alps are all from Permo-Triassic and Early Tertiary rocks, apart from some recently added Cretaceous data of the Vicentinian Alps (see List 2). On the other hand no comparable data have been obtained so far from the Permo-Triassic of the Apennines.

We therefore decided to study paleomagnetically the Jurassic—Cretaceous

LIST 1

Paleomagnetic studies carried out in the Northern Apennines

Localities	Age covered by the samples	References
A	Late Camp. to Early Eoc.	Premoli Silva et al., 1974
A	Cenomanian to Eocene	Lowrie and Alvarez, 1974,1975
A	Cenomanian to Eocene	Channell and Tarling, 1975
B	L. Jurassic to Eocene	Klootwijk and VandenBerg, 1975
B	M/L. Jurassic to Eocene	VandenBerg et al., 1976
C	Cenomanian to Eocene	Lowrie and Alvarez, 1974, 1975
C	Cenomanian to Eocene	Channell and Tarling, 1975
D	L. Carboniferous to E. Permian and M/L. Triassic	Present paper

Letters refer to Fig. 1, South of the Po-basin.

limestone sequence of the Vicentinian Alps and the Verrucano of Tuscany in the Northern Apennines.

In this paper the first results of this attempt are presented and a tentative interpretation is given.

PREVIOUS STUDIES, GEOLOGY AND SAMPLING LOCALITIES

South of the Po-basin

Paleomagnetic data from areas South of the Po plain are available only

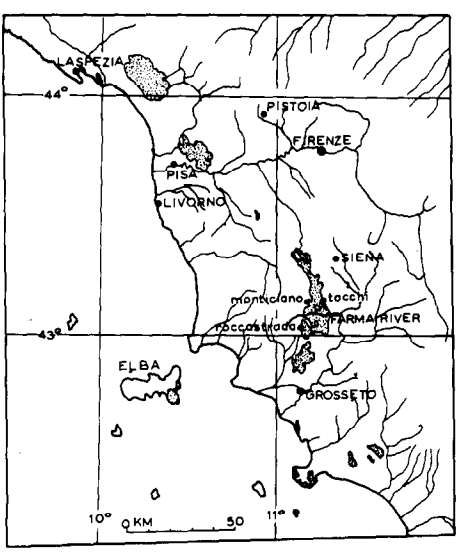


Fig. 2. Outcrops of the autochthonous basement (dotted: Verrucano) in Tuscany, after Dallan Nardi and Nardi (1972).

from the Umbrian sequence and cover the Early/Middle Jurassic to Eocene timespan (see List 1). For a discussion of the results we refer to an other paper (VandenBerg et al., 1976).

In Fig. 2, the outcrops of the autochthonous basement in the Northern Apennines are indicated. An important problem in finding appropriate sampling localities is the metamorphism that has affected large parts of the Tuscan autochthon. This low-grade metamorphism was dated 11—14 m.y. B.P. (Giglia and Radicati, 1970). Sites must be found where the primary paleomagnetic directions have not been eliminated by remagnetization.

The non-metamorphic rocks of Late Triassic to Tertiary age of Central Tuscany, farther to the East, are supposed to be in allochthonous positions. Therefore they are not especially suitable for paleomagnetic research either.

The grade of metamorphism of the autochthonous basement, however, varies from one place to another. Field study of the outcrops showed that the area between Grosseto and Siena was least affected. Another positive point in favour of this area is its simple tectonic structure, compared with other parts. The basement in the area near Roccastrada (Fig. 2) was only involved in minor folding and Pliocene normal faulting (Signorini, 1966).

The geological setting of the section is as follows: Folded dark-grey sandstones and dark shales of the Hercynian cycle (Farma Formation), containing Middle Carboniferous to Early Permian floras and faunas, and correlable with the Buti phyllites and quartzites 275 m.y. old (Bortolotti et al., 1970; Borsi et al., 1966), are unconformably overlain by Verrucano rocks. No direct age determinations of this Verrucano are known. A minimum age limit is given by the overlying Noric to Rhaetic Grezzoni and Calcare Cavernoso Formations (Bortolotti et al., 1970; Dallan Nardi and Nardi, 1972).

Several authors have pointed to the striking resemblance in facies and stratigraphic position between this "Siena" Verrucano and the Verrucano of the Alpi Apuane, which was dated as Ladinian to Carnian (Rau and Tongiorgi, 1968; Gelmini, 1969; Gianni et al., 1972; Gasperi and Gelmini, 1973). Provisionally, we assume this age for the "Siena" Verrucano.

Along the road from Monticiano to Tocchi (Fig. 2), the "Siena" Verrucano is exposed in its typical facies, containing polygenic conglomerates, quartzitic layers, grey and violet-red sandstones, grey siltstones and green phyllitic shales with sandstone and quartzite intercalations.

We restricted the sampling (8 sites) to fine-grained, grey and violet-red sandstones without visible crossbedding. Samples at one additional site were collected from rocks fitting this description for the Farma Formation (Bortolotti et al., 1970).

North of the Po-basin

In List 2 the previous work is compiled, and we refer to Zijderveld and Van der Voo (1973) for extensive discussion of the Permo-Triassic data.

Encouraged by the good paleomagnetic results we have obtained from red

LIST 2

Paleomagnetic studies carried out in the Southern Alps

Localities	Age covered by the samples	References
A	Eocene/Oligocene	Soffel, 1972, 1974
B	Eocene/Oligocene	Soffel, 1975a, 1975b
B	Eocene and Oligocene	De Boer, 1963, 1965
C	Upper Cretaceous	Channell and Tarling, 1975
D	Jurassic and Upper Cretaceous	Present paper
E	M/L. Permian and E/M. Triassic	De Boer, 1963, 1965
F	M/L. Triassic	Manzoni, 1970
F	M/L. Permian	Manzoni, 1970; Guicherit, 1964
F	E. Permian	Zijderveld et al., 1970; Van Hilten, 1960, 1962
G	M. Triassic and E. Permian	Zijderveld and De Jong, 1969

Letters refer to Fig. 1, North of the Po-basin.

pelagic limestones, we searched for such deposits in the Southern Alps. For several reasons the Late Mesozoic to Eocene limestone sequence of the Vicentinian Alps was regarded to be the preferable target. This sequence has a stratigraphy nearly identical to that of the Umbrian sequence in the Northern Apennines, of which it seems to be the continuation (Aubouin, 1965). Tectonic disturbance is confined to normal faulting and folding along the

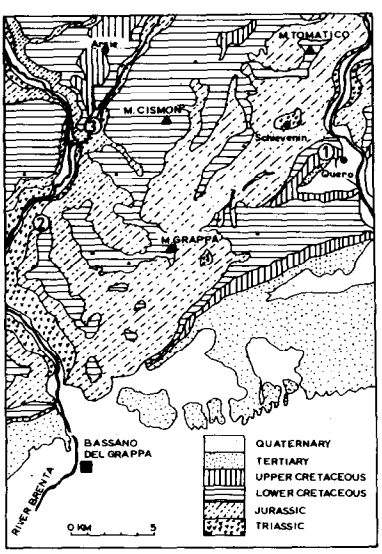


Fig. 3. Sketchmap of the outcropping rocks North of Bassano del Grappa (Fig. 1) with the sampling localities referred to in the text.

Northern margin of the Po-basin related to its foundering. North of this margin, the sediments are in subhorizontal position, overlying the thick Triassic dolomites, which, in turn, have an undisturbed stratigraphic contact with underlying older rocks.

Since previous Late Mesozoic data were confined to the Upper Cretaceous, and were not accurately dated, we directed our research to the Jurassic and Upper Cretaceous limestones. Several suitable levels in this sequence were found. At two different levels, the limestones are developed in the "Ammonitico Rosso" facies, one of Kimmeridgian, the other of Tithonian age. Both were sampled (localities 2 and 3 in Fig. 3). The Lower Cretaceous rocks consist of white and green, well stratified limestones with shaly intercalations, which are sometimes dark and bituminous. The Upper Cretaceous is exposed as white (Cenomanian) and pink to red (Turonian to Maastrichtian), well stratified, pelagic limestones, containing abundant planktonic foraminiferal faunas. The latter were sampled in a continuous section along the road from Quero to Schievenin (locality 1 in Fig. 3). All samples could be dated accurately by studying the planktonic foraminifera in thin sections. The relative ages of the faunas were established by application of Postuma's zonation (1971).

Although the number of sites is still limited and further research is needed (the study is being continued), a record of dated paleomagnetic directions is now achieved and comparison with data from more southern areas is possible.

DEMAGNETIZATION DATA

In the paleomagnetic laboratory of the State University of Utrecht core specimens (2.2 cm height, 2.5 cm diameter) were drilled from the hand-samples. The measurements were carried out on the very sensitive Czechoslovakian JR-3 spinner. The specimens were subjected to standard alternating fields for separation of the various magnetization components. No thermal cleaning was applied since these standard AF demagnetizations turned out to be most useful and adequate. Pilot specimens (at least one of each site) were progressively demagnetized in 10—15 steps up to the maximum obtainable peak value of 3100 Oe. According to the experiences with these pilot demagnetizations all specimens were partially demagnetized in 8—10 steps in the appropriate AF range. The characteristic and secondary magnetization components could be separated and their directions determined with the useful help of orthogonal projection figures (Zijderveld, 1967, 1975) which show sequential changes of the magnetization vector upon demagnetization.

The Mesozoic limestone sequence of the Vicentinian Alps

Fifty samples were collected from ten sites in this sequence. The initial intensities of remanent magnetization appeared to be low: 10^{-5}—10^{-6}

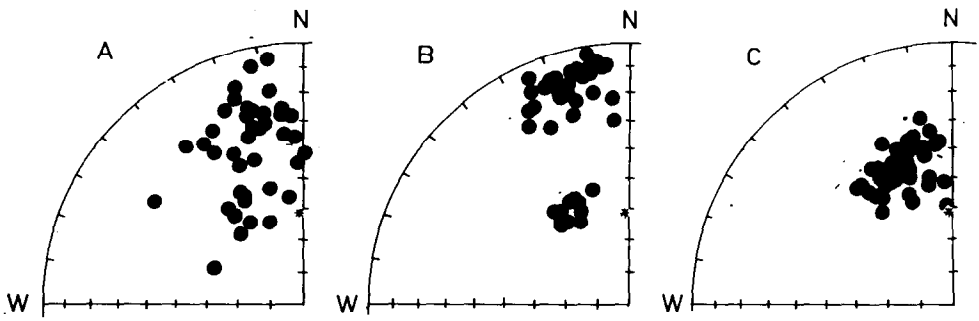


Fig. 4. Remanent magnetization directions in equal-area projection (of samples from the Vicentinian Alps), pointing downward in lower hemisphere projection. Asterisk denotes present local field direction. A. Initial N.R.M. sample directions. B. Characteristic sample directions, after demagnetization and before dip correction. C. Same, after dip correction.

Gauss, which is normal for pelagic limestones. The initial sample directions were distributed over the lower hemisphere (Fig. 4A) in the northwestern quadrant only. Three components were discerned during demagnetization. The first remanent magnetization component was mostly removed from 0 to 200 Oe peak value, clearly related to the present local field (359°, +62°) and therefore of secondary origin. The supposedly characteristic component could be removed from 150 to 500 Oe peak value. A third very hard magnetization component was sometimes observed and could be removed from 500 to 3000 Oe peak value; being related to the present local field it too is therefore assumed to be of secondary origin.

The samples of two sites (BKG and TIA) contained only unstable remanence and revealed directions scattered over the lower hemisphere. Both sites were omitted from further considerations.

Five orthogonal projection figures are shown in Fig. 5, which show the characteristics of demagnetization. Figs. 5A,B and C show the removal of a secondary component in the so-called "Scaglia" (Turonian to Maastrichtian) samples from 0 to 150 Oe peak value. The characteristic magnetization component, having 60—80% of the initial intensity, is removed upon further demagnetization, with the magnetization vector path going through the origin. The removal of the magnetization components in Fig. 5D is similar; this specimen came from a Rosso Ammonitico sample. In Fig. 5E a secondary component is removed from 0 to 55 Oe peak value, and the characteristic component is removed in the 55—300 Oe peak value range. From 300 to 1000 Oe peak value a third secondary component is removed, having a direction close to the present field. Other specimens from the same site (DIB) had demagnetization curves similar to that shown in Fig. 5D. The same phenomenon was observed in some samples of the Umbrian sequence (VandenBerg et al., 1976).

In Fig. 4B the characteristic sample directions are plotted. The directions are clearly concentrated in two distinct groups, which is due to a different

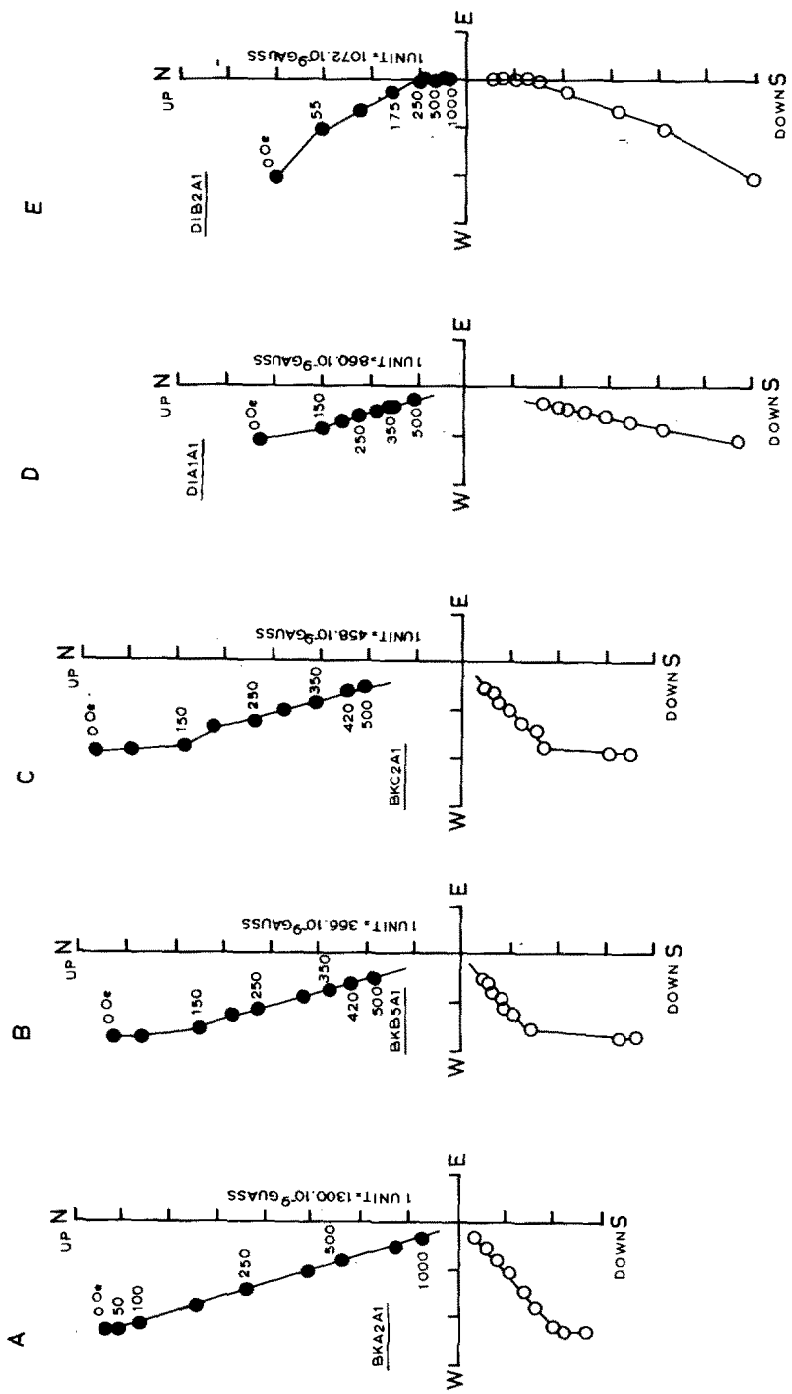


Fig. 5. Orthogonal projection figures of the vector path of the total remanent magnetization vector during AF cleaning of samples from the Vicentinian Alps. Circles are projections on the vertical East—West plane, dots are projections on the horizontal plane. Numbers denote Oersted peak values of the applied alternating fields.

TABLE I

Remanent magnetization parameters, measured in samples from the Vicentinian Alps

Sites	Samples	Initial intens. (10^{-8} Gauss)	Mean direction *	α_{95} (deg.)	Age	Mean direction ** (deg.)	α_{95} (deg.)	Pole position (deg.)	*dp* (deg.)	*dm*
BKG	5	52—104	unstable	—	Maastrichtian	—	—	—	—	
BKF	5	86—140	339.5 + 49.8	6.8	Late Santonian /Early Camp.	342.7 + 50.9	5.3	70.4 N 119.1 W	4.8	7.1
BKE	5	68—105	346.1 + 51.9	9.9						
BKD	5	142—768	337.6 + 44.8	8.6	Early Santonian	344.2 + 43.3	5.3	65.7 N 131.3 W	4.1	6.6
BKC	5	287—734	350.4 + 41.5	5.5						
BKB	5	275—740	343.7 + 39.4	3.8	Late Turonian /Coniacian	345.5 + 38.2	3.3	62.7 N 137.5 W	2.3	3.9
BKA	5	837—1265	347.3 + 36.8	6.3						
TIA	5	59—171	unstable	—	Tithonian	—	—	—	—	
DIA	5	475—950	328.3 + 44.0	5.5	Kimmeridgian	328.5 + 45.3	3.0	58.5 N 104.7 W	2.4	3.8
DIB	5	135—805	328.8 + 46.5	4.2						

* After cleaning and dip correction. ** $N = 10$.

bedding tilt of the samples concerned. The unfolding is seen to be clearly positive in Fig. 4C, where all characteristic sample directions are plotted having been corrected for the bedding tilts. For computation of the mean directions (Table I) Fisherian statistics were used. The overall mean direction of the Cretaceous samples is in good agreement with Channell and Tarling's (1975) result (List 2).

The Tuscan Verrucano

46 samples were collected at nine sites throughout the "Siena" Verrucano. The initial intensities ranged from 3.10^{-5} to 1.10^{-6} Gauss, which is rather low. In Fig. 6A the initial sample directions are plotted in an equal-area projection. Only normal polarities were observed. The directions show a reasonable scatter with a preference for the northwestern quadrant.

During AF demagnetization of most specimens a present-day local field direction (359°, +59°) appeared to be present, but in the samples of sites VUE, VUF, VUG and VUK this component was of minor importance. In the case of sites VUC, VUH and VUB a rather rigorous cleaning was necessary to determine the characteristic magnetization components. Some examples of demagnetizations are given in Fig. 7. In Fig. 7A it is shown that a present local field component is removed from 0 to 500 Oe peak value and afterwards the harder (characteristic) second component is broken down in higher fields. In Fig. 7D it is apparent that AF demagnetization results only in a minor change in direction of the magnetization component. In the case of Fig. 7B it is clear that, although the declination varies very little from 0 to 2500 Oe peak value demagnetization, the inclination reaches its characteristic value only in AF fields above 2500 Oe. Site VUC (Fig. 7B) was taken from a violet-red sandstone, as was site VUH (Fig. 7C). In Fig. 7C it is clear that the speci-

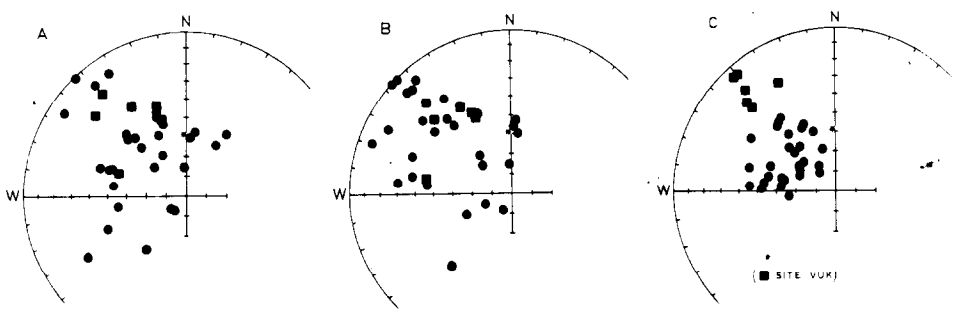


Fig. 6. Directions of remanent magnetization in equal-area projection (of samples from the Apennines), all directions pointing downward in lower hemisphere projection. A. Initial N.R.M. sample directions. B. Characteristic sample directions after demagnetization and before dip correction. C. Idem, after dip correction. Squares represent sample directions of Site VUK.

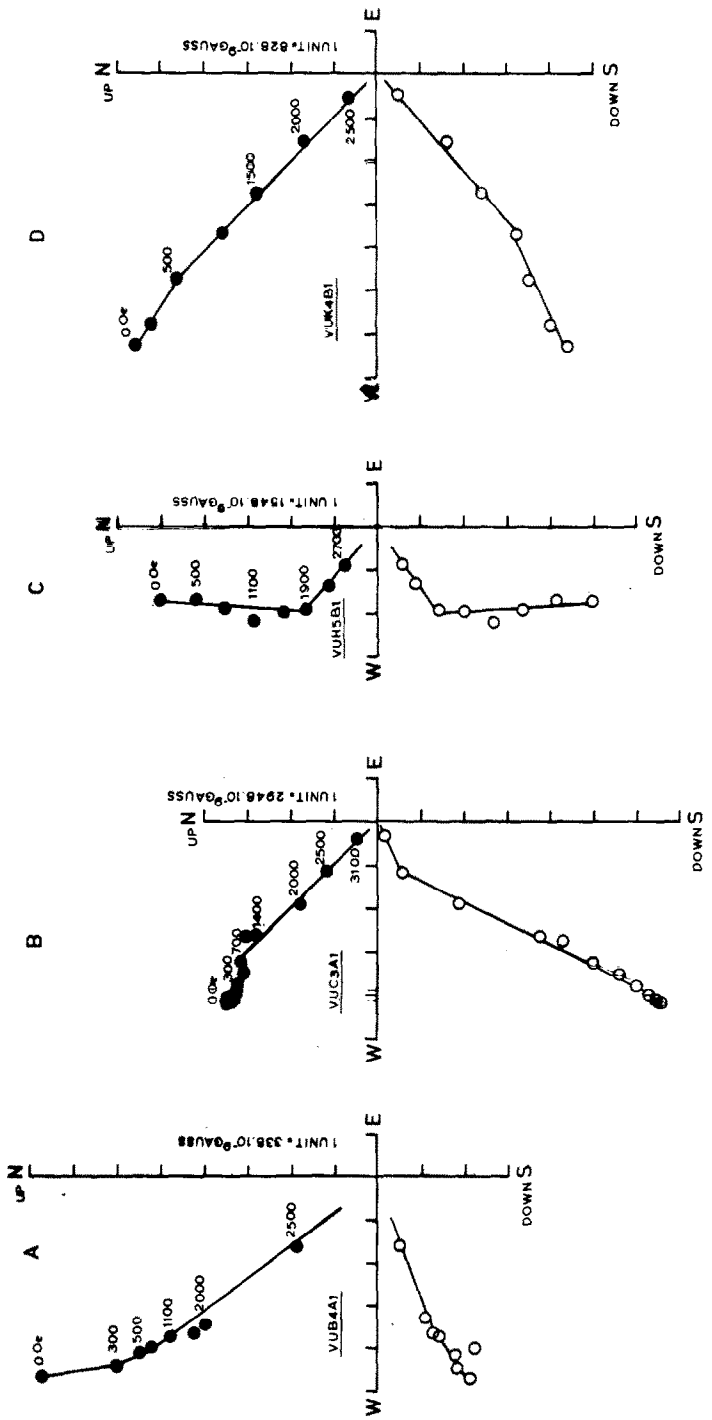


Fig. 7. Orthogonal projection figures of the vector path of the total remanent magnetization vector during AF cleaning, for samples from the Apennines.

TABLE II

Remanent magnetization parameters, measured in samples from the Apennines

Sites	Samples	Initial intens. (10^{-8} Gauss)	Mean direction *	α_{95} (deg.)	
VUA	5	6—12	too weak	—	
VUB	5	118—337	327.2 + 55.7	11.6	
VUC	3	1808—2086	314.1 + 53.1	26.3	
VUD	5	1642—1875	unstable	—	Mean direction for sites VUB,C,D,E,F,G,H:
VUE	5	310—459	295.7 + 63.8	12.1	301.1 + 60.2, α_{95} = 6.0, N = 33
VUF	5	211—319	283.9 + 47.5	14.5	Pole position : 47.1 N 59.6 W (dp = 6.9°, dm = 9.1°)
VUG	5	124—212	288.1 + 68.0	20.5	
VUH	5	655—1126	300.1 + 64.8	10.6	
VUK	6	524—786	318.8 + 20.8	12.8	Pole position : 41.9 N 107.7 W (dp = 7.1°, dm = 13.5°)

* After cleaning and dip correction.

men from this site (VUH) has a notable present local field component, which is removed from 0 to 2000 Oe peak value.

Site VUA turned out to be too weak to reveal reliable components of magnetization during AF cleaning, and was therefore omitted. Site VUD appeared to be unstable during demagnetization. Two samples of site VUC could not be demagnetized properly in alternating fields up to the maximum obtainable peak value of 3100 Oe.

In Fig. 6B the sample directions are plotted which are assumed to be characteristic since removal of present field components was adequate, and these directions exhibit a positive fold test.

In Fig. 6C these characteristic sample directions are plotted corrected for dip. The scatter is clearly diminished. The results from site VUK (Table II) differ notably from the other results. The samples of site VUK were taken from an anticlinally folded sandstone bed and all had different dips; the fold test is definitely positive. The lower inclination may well be due to the fact that VUK was taken from Late Carboniferous to Early Permian rocks of the Farma formation, while the other sites have an appreciably younger age (Ladinian to Carnian). In Table II the results are given with application of Fisherian statistics.

INTERPRETATION

North of the Po-basin

All data from the Vicentinian Alps (List 2 and Table I) are given in equal-area projection in Fig. 8A. The data from the other areas in the Southern Alps have not been plotted to avoid overcrowding. Zijderveld and Van der Voo (1973) showed that these do not differ significantly from each other. Figure 8A can be considered as representative for the Vicentinian Alps. In the following the paleomagnetic directions are interpreted as movements with respect to the pole.

The Early to Middle Triassic directions do not differ from the Jurassic (Kimmeridgian) result (Fig. 8A), which means that no paleomagnetically detectable movements took place in the corresponding time span. A southward movement of about 10° of latitude is concluded from a correspondingly lower inclination value in the Cenomanian (Channell and Tarling, 1975). This movement may be correlated with the Late Aptian to Early Cenomanian southward movement phase in the Northern Apennines (VandenBerg et al., 1975, 1976). A 15° counterclockwise rotation of Late Cenomanian to Early Turonian age, followed by a northward movement of about 10° of latitude in Santonian to Early Campanian times, is deduced from the change of paleomagnetic directions in this time interval. A further post-Cretaceous counterclockwise rotation (15°) and an additional northward movement is suggested, since Late Cretaceous as well as Eocene data are still different from the present local field direction (see Fig. 8A). The paleomagnetic directions in

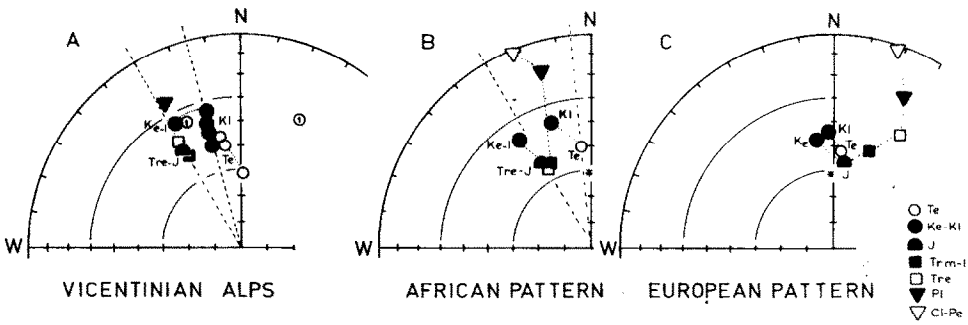


Fig. 8. Comparison of the mean directions from the Vicentinian Alps with African and European data; equal-area lower hemisphere projection. Asterisk denotes the present-day local field direction. All directions are pointing downward.
A. The actually measured directions from the Vicentinian Alps (List 2). Data from the Monte Lessini Volcanics (Soffel, 1975): ①
B. In this projection the Vicentinian Alps are considered as fixed to the African plate. The directions to be expected were computed from the poles for Africa of Van der Voo and French (1974) for Paleozoic and Jurassic to Early Tertiary, and from the Triassic poles of McElhinny and Brock (1975).
C. The Vicentinian Alps are considered in this projection as fixed to the European plate and the directions to be expected were computed from the poles of Van der Voo and French (1974).

Oligocene rocks (De Boer, 1963, 1965) are very close to the present local field direction, indicating that these ultimate latitudinal and rotational movements are of Late Eocene to Oligocene age. The results of Soffel (1975a, 1975b) from the Monte Berici and Lessini (see Fig. 8A) are interpreted as showing this final Late Eocene to Oligocene rotation. These data have comparatively aberrant low inclination values, which may be due to unidentified tectonic complications and/or to errors in the dip corrections. The mean result from the Eocene—Oligocene Colli Euganei (Soffel, 1972) is in good agreement with De Boer's findings (1963, 1965).

In Figs. 8B and 8C those paleomagnetic directions are plotted, which would be expected in the Southern Alps, fixed with respect to either the African plate (Fig. 8B) or the European plate (Fig. 8C). In Fig. 8B these directions were computed from the paleomagnetic poles proposed for Africa by Van der Voo and French (1974) for the Paleozoic, Jurassic, Cretaceous and Early Tertiary. In Fig. 8C their paleomagnetic European poles were used for Paleozoic, Mesozoic and Early Tertiary times. These African and European paleomagnetic poles were deduced from the actually measured paleomagnetic poles of the Atlantic-bordering continents, and from the reconstruction of the sea-floor spreading history of the Atlantic (Pitman and Talwani, 1972). In Fig. 8B we used the Triassic African paleomagnetic poles published by McElhinny and Brock (1975), since these seem to have superseded Van der Voo and French's poles (1974) for this period.

It can be seen at a glance, that the Vicentinian (Southern) Alps belonged

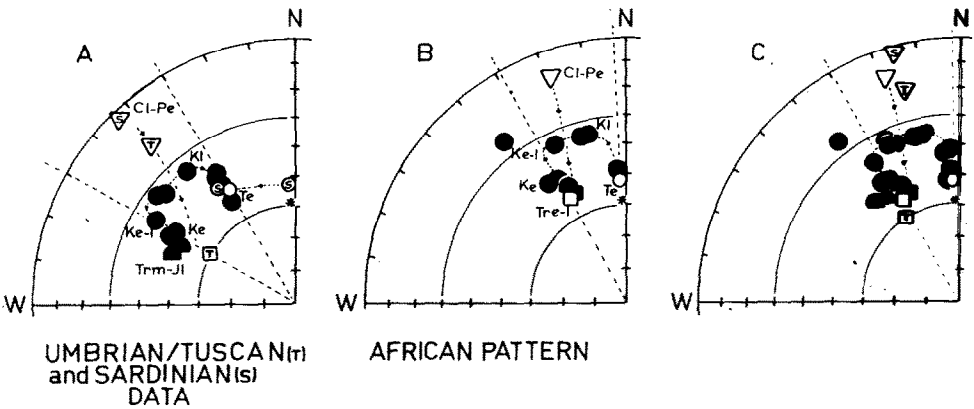


Fig. 9. Comparison of the mean directions from the Northern Apennines with the African data. Equal-area lower hemisphere projection. Asterisk denotes local field direction in the Siena area.
A. The actually measured directions from the Northern Apennines (VandenBerg et al., 1976) with the Tuscan (T) and the Sardinian data (S) added (Manzoni, 1974; present paper).
B. In this projection the Northern Apennines are considered as a part of the African plate and the directions to be expected were computed from the actually measured African poles (McElhinny and Brock, 1975). See Fig. 8B for comparison.
C. In this projection the Northern Apennines' pattern has been accounted for a 25—30° rotation relative to Africa. The Sardinian data are rotated over the same amount for comparison.

to the African plate rather than to the European plate. However, two important discrepancies can be observed. The African plate rotated during the entire Late Cretaceous about 30°, while the Vicentinian Alps rotated only 15°. The Early Tertiary rotation of the Vicentinian Alps was independent of and relative to the African plate. The Late Cretaceous (about 15°) and the Early Tertiary (about 15°) rotations bring the total rotation amount for the Vicentinian Alps up to about 30°, relative to the pole. This explains the similarity of the Late Paleozoic data from the Southern Alps and Africa (Zijderveld and Van der Voo, 1973). Timing and amounts of the south- and northward movements of the Vicentinian Alps and Africa is nevertheless the same (within the accuracy of this study).

South of the Po-basin

In Fig. 9A the Northern Apennines pattern is plotted. For computation of the paleomagnetic directions, the data from Umbria of VandenBerg et al. (1976) were used, since these are carefully dated by a study of the planktonic foraminifera and cover the longest timespan (List 1). For comparison the Sardinian data (compiled by Manzoni, 1974) were averaged for the Permian, Tertiary and Quaternary, and plotted in Fig. 9A as well. In Fig. 9B the ac-

tually measured African directions for the Siena area in its present position are plotted, computed from the paleomagnetic poles compiled by McElhinny and Brock (1975). Comparison with Fig. 8B shows clearly the overall identity with this African pattern and that according to the simulated paleomagnetic poles of Van der Voo and French (1974). This firmly supports their basic assumption that the movements of the African plate were controlled by the opening history of the Atlantic (according to Pitman and Talwani, 1972). The following history of movements with respect to the pole is deduced for the Northern Apennines: A southward movement took place in Albian to Early Cenomanian times, followed by a counterclockwise rotation of 30° during Turonian to Late Santonian/Early Campanian. Meanwhile a northward movement occurred in Santonian to Campanian. Campanian to Early Eocene data show no changes and still differ about 30° from the present local field direction. This indicates that a post-Early Eocene rotation must have taken place. Since the deduced movement pattern of the Northern Apennines and the African movement pattern (Figs. 9A and B) show a close identity, apart from the post-Early Eocene rotation, it is concluded that the Northern Apennines were part of the African plate until the Early Tertiary, when an independent counterclockwise rotation started relative to the African plate. After accounting for this rotation (25—30°) both patterns cover each other completely, as shown in Fig. 9C (VandenBerg et al., 1975, 1976).

In Fig. 9C it is shown that the newly obtained data from Tuscany are in very good agreement with the comparable African data. This rules out the possibility that the Tertiary rotation, shown for the Umbrian sequence, was caused by a décollement (Lowrie and Alvarez, 1975) of this sequence only.

The Sardinian data are in excellent agreement with the Northern Apennines' pattern (Fig. 9A) and with the African pattern (Fig. 9C) after accounting for a 25—30° rotation. This suggests that Sardinia belonged all the time to the Italian Peninsula. This raises interesting consequences for our understanding of the origin of the Tyrrhenian Sea.

The fault zone underneath the Po-basin

Comparison of paleomagnetic data from crustal blocks North (Fig. 8A) and South (Fig. 9A) of the Po-basin, covering a long timespan, is now possible. It has been shown that both blocks belonged to the African plate and that their movements were mostly defined by the opening history of the Atlantic (Fig. 8A-8B, Fig. 9A-9B). Differences between both movement patterns must have resulted in a fault zone in the area of the Po-basin, as argued below.

The first paleomagnetically observed difference in movements occurred in the Late Cretaceous. Both crustal blocks together with Africa started the Late Cretaceous rotation in the Turonian/Cenomanian, but the Vicentinian Alps did not continue through 30°, as the Northern Apennines and Africa did, but stopped after 15°. The Northern Apennines continued to rotate 15°

during the Coniacian to the Late Santonian/Early Campanian, while the Vicentinian Alps stopped the rotation in the Late Turonian. Nevertheless, both crustal blocks moved Northward with the African plate during the Santonian to Campanian. The reason why the northernmost extension of the African plate did not rotate another 15° can possibly be sought in a Late Cretaceous collision with parts of the European plate.

The second observed difference occurred in the Early Tertiary (Late Eocene to Oligocene). The Northern Apennines and the Vicentinian Alps were subjected to an independent counterclockwise rotation relative to Africa of about 25° for the first and about 15° for the latter. The volcanic activity in the Southernmost Alps is probably related to the relative movements involved.

The precise nature of the relative movements can only be surmised since the rotation-poles are not known. No significant difference in inclinations is observed, but North—South movements of less than 400 km fall within the accuracy of our paleomagnetic data and are therefore not detectable.

CONCLUSIONS

(1) The Vincentinian Alps (possibly all of the Southern Alps) belonged to the African plate until the Late Turonian. Afterwards the "Vicentinian crustal block" was partially uncoupled from the African plate as well as from the Italian Peninsula.

(2) The Northern Apennines (possibly the whole Italian peninsula + Sardinia) were part of the African plate until the Early Tertiary when the "Northern Apennines crustal block" was partly uncoupled from the African plate by a counterclockwise rotational movement.

(3) During the Late Cretaceous (Coniacian to Late Santonian/Early Campanian) and again in Early Tertiary (Late Eocene to Oligocene) a fault zone must have been active in the area of the Po-basin. This supports Mueller and Talwani's hypothesis (1971). Thus the Po-basin can be seen as a fault zone, buried under a thick Late Tertiary to Quaternary cover.

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